

Contribution submission to the conference Erlangen 2026

Status of large-area cryogenic microcalorimeter (LAMCAL) development for DELight — •LENA HAUSWALD¹, CHRISTIAN JEUP¹, FRIEDRICH WAGNER¹, and SEBASTIAN KEMPF^{1,2} — ¹Institute of Micro- and Nanoelectronic Systems, Karlsruhe Institute of Technology (KIT), Germany. — ²Institute for Data Processing and Electronics, Karlsruhe Institute of Technology (KIT), Germany.

The Direct Search Experiment for Light Dark Matter (DELight) is a pioneering initiative aimed at probing light dark matter (LDM). By utilizing superfluid ^4He as target material, DELight will enable the exploration of weakly interacting DM candidates with masses notably below $100 \text{ MeV}/c^2$. To achieve the required detector sensitivity for resolving the various signal channels of the superfluid, both in terms of time and energy resolution, DELight focuses on the development of large-area cryogenic microcalorimeters (LAMCALs) based on magnetic microcalorimeter (MMC) technology.

In contrast to state-of-the-art MMCs, each LAMCAL will be operated as athermal phonon detector. Here, the energy of athermal phonons generated by a energy deposit and propagating ballistically within the absorber is converted into a temperature rise of the paramagnetic temperature sensor via Al-based superconducting phonon collectors. Consequently, the signal rise time is governed by the phonon collection time, while the absorber heat capacity becomes negligible to first order, enabling excellent energy resolution. This contribution presents the current status of LAMCAL development for DELight, highlighting their potential and the challenges encountered to date.

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